

Economic analysis of a genomic report

Cost–utility model of acting on the findings in one whole-genome input, valued at \$100,000 per QALY from a healthcare sector perspective.

Source input	synthetic_wgs.vcf	Input type	WGS
Generated	05 September 2026	Time horizon	10 years
Discounting	3.0%, costs and QALYs	Build	aa929da 2026-09-05T10:07:10-04:00

Synthetic input — no human genome was used.

Every finding in this report was produced from a purpose-built synthetic genome generated by this repository. No personal genomic or health data was read, stored or published to produce it.

Research and educational use — not medical or financial advice.

This is a decision-analytic model, not a clinical or economic evaluation submitted to any authority. Its parameters are population averages applied to one genome: it values the expected consequences of acting on a pattern of findings, and does not predict what will happen to any individual. Monetary figures include health valued at a willingness-to-pay threshold and are not cash returned to anyone. Clinically relevant findings require confirmation by an appropriate professional — a physician, clinical pharmacist or board-certified genetic counsellor — before anything follows from them.

Findings-first economics

Acting on these findings is modelled to improve health and reduce healthcare spending.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

HEALTH GAIN

~23 days

0.063 incremental QALYs

of healthy life, on average, for a person with this pattern of findings

HEALTHCARE COST CHANGE

-\$27,378

healthcare-sector perspective

negative means the modelled programme spends less than usual care

MODELLED NET BENEFIT

\$33,637

reference-case NMB

at \$100,000 per QALY. This values health at the threshold; it is not cash.

DECISION UNCERTAINTY

1,500 of 1,500

modelled simulations cost-effective (100%); cost-saving in all 1,500 of 1,500 simulations (100%) · 95% interval \$14,123 to \$64,129 across 62 varied parameters

19

findings carry a standardised economic estimate

6 high-confidence, 6 moderate-confidence, 7 low-confidence. 13 of the 32 findings shown are reported without a dollar figure rather than being assigned one the model cannot support.

BRCA1 pathogenic variant

LDLR familial hypercholesterolemia

APOB familial hypercholesterolemia

CHEK2 I157T carrier

APOE ε4 carrier

PTPN22 R620W carrier

Factor V Leiden / Prothrombin...

OPRM1 OPRM1 A118G (mu-opioid...

LRRK2 G2019S carrier

CYP2C19 Intermediate Metabolizer

G6PD deficiency

DRD2/ANKK1 DRD2 / ANKK1 Taq1A

CYP2D6 Intermediate Metabolizer

NAT2 — NAT2 slow acetylator —...

PON1 — PON1 Q192R —...

SLCO1B1 Intermediate Function

ATP7B variant

Coeliac-permissive HLA haplotype...

Vitamin D Synthesis Efficiency

2 actionable wellness variant(s)

HFE C282Y carrier

GABRA2

and 10 more

What rises to the top

Ranked by expected net monetary benefit, each valued on its own.
Finding-level values are not additive; the pooled reference case is calculated separately.

BRCA1 PATHOGENIC VARIANT

\$17,308

Enhanced screening + risk-reducing surgery option

HIGH

LDLR FAMILIAL HYPERCHOLESTEROLEMIA

\$9,442

High-intensity statin + cascade screening

HIGH

APOB FAMILIAL HYPERCHOLESTEROLEMIA

\$6,658

High-intensity statin therapy

HIGH

CHEK2 I157T CARRIER

\$3,739

Risk-aware screening schedule

MODERATE

Your findings at a glance

Findings that could change a decision

Every finding the model examined, the decision it bears on, the strength of the evidence, and what it is worth on its own. No dollar figure appears without the genomic result that produced it — the detail behind each figure is on the pages that follow.

MEDICATION & PRESCRIBING 3 findings

CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)	Clopidogrel prescribing	HIGH	\$490
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)	Codeine prescribing	HIGH	\$210
SLCO1B1 Intermediate Function (IM-like) (simvastatin)	Simvastatin prescribing	HIGH	\$90

RISK & PREVENTION 9 findings

BRCA1 pathogenic variant (heterozygous)	Enhanced screening + risk-reducing surgery...	HIGH	\$17,308
LDLR familial hypercholesterolemia (heterozygous)	High-intensity statin + cascade screening	HIGH	\$9,442
APOB familial hypercholesterolemia (heterozygous)	High-intensity statin therapy	HIGH	\$6,658
CHEK2 I157T carrier — moderate cancer risk awareness	Risk-aware screening schedule	MODERATE	\$3,739
APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)	Intensive cardiovascular risk reduction...	MODERATE	\$3,514
PTPN22 R620W carrier — autoimmune screening awareness	Awareness for autoimmune symptom clusters	MODERATE	\$2,380
Factor V Leiden / Prothrombin carrier — situational anticoagulation	Awareness for high-risk situations	MODERATE	\$2,273
G6PD deficiency — oxidative drug avoidance	Avoid primaquine/dapsone/rasburicase...	MODERATE	\$463

Findings that could change a decision, continued

RISK & PREVENTION (CONTINUED) 1 of 9 findings · 8 shown earlier

Coeliac-permissive HLA haplotype (DQ2.5/DQ8) — serology decision	tTG-IgA serology if symptomatic, before any...	MODERATE	-\$122
--	--	----------	--------

LOWER-CONFIDENCE & EXPLORATORY 8 findings

OPRM1 OPRM1 A118G (mu-opioid receptor) — OPRM1 A118G...	OPRM1-guided analgesic dosing + AUD treatment...	LOW	\$699
LRRK2 G2019S carrier — Parkinson's early intervention	Prodromal PD monitoring + neuroprotective...	LOW	\$613
DRD2/ANKK1 DRD2 / ANKK1 Taq1A — DRD2/ANKK1 Taq1A — reward-pathway...	Dopamine-aware addiction treatment +...	LOW	\$442
NAT2 — NAT2 slow acetylator — occupational cancer risk awareness	Aromatic amine exposure monitoring + bladder...	LOW	\$90
PON1 — PON1 Q192R — organophosphate susceptibility	Pesticide exposure avoidance + cholinesterase...	LOW	\$90
ATP7B variant — Wilson's disease copper monitoring	Serum copper/ceruloplasmin monitoring + early...	LOW	-\$9
Vitamin D Synthesis Efficiency — Nutrient-metabolism variant...	Targeted supplementation	LOW	-\$216
2 actionable wellness variant(s)	Nutrient metabolism, sleep, and fitness...	LOW	Not yet standardised

REPORTED, NOT COSTED 12 findings

HFE C282Y carrier — reproductive awareness	Partner testing for reproductive risk...	NO GRADE	Not routed to the engine
GABRA2 — Alcohol-metabolism susceptibility variant	Personalized AUD risk awareness +...	NO GRADE	Not routed to the engine

Findings that could change a decision, continued

REPORTED, NOT COSTED (CONTINUED) 10 of 12 findings · 2 shown earlier

OPRM1 — Opioid-response pharmacogenomic variant	OPRM1-guided opioid dosing + naltrexone...	NO GRADE	Not routed to the engine
CHRNA5 — Nicotine-dependence susceptibility variant	CHRNA5-guided smoking cessation	NO GRADE	Not routed to the engine
CRHR1 — Stress-axis variant (CRHR1/FKBP5)	Trauma-informed care + HPA-axis-targeted...	NO GRADE	Not routed to the engine
CYP2C19 — top-drug actionable (amitriptyline)	Genotype-guided prescribing decision for...	NO GRADE	Not routed to the engine
CYP2D6 — top-drug actionable (atomoxetine)	Genotype-guided prescribing decision for...	NO GRADE	Not routed to the engine
SLCO1B1 — top-drug actionable (atorvastatin)	Genotype-guided prescribing decision for...	NO GRADE	Not routed to the engine
CYP2C9 — top-drug actionable (fluvastatin)	Genotype-guided prescribing decision for...	NO GRADE	Not routed to the engine
Reproductive genetics — 12 carrier condition(s) where partner testing...	Consider partner carrier testing and genetic...	NO GRADE	Not routed to the engine
HBB carrier — heterozygote health	Carrier Screening	NO GRADE	Not routed to the engine
SERPINA1 carrier — heterozygote health	Carrier Screening	NO GRADE	Not routed to the engine

Finding-level values are standalone expected values. They are not additive. The reference-case total is calculated separately after overlapping signals are pooled by condition.

Expected NMB values health at \$100,000 per QALY and is **not cash returned to anyone**. Ordering is by clinical group first and expected NMB second.

13 findings are shown without a standardised expected NMB rather than being assigned a value the model cannot support.

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)

clopidogrel

WHAT WAS FOUND

Genotype predicts altered handling of clopidogrel.

WHY IT MATTERS

Clopidogrel is a prodrug that CYP2C19 must activate. Reduced activity means less active drug and a higher chance the antiplatelet effect is inadequate.

WHAT COULD CHANGE

Avoid clopidogrel non-response / stent thrombosis (MACE)

KEY DEPENDENCY

Value accrues only if clopidogrel is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

\$490

Medical cost averted	\$300
Health gain	0.002 QALY
Cost to act	\$10 registry default

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on the methods page.

CYP2D6 Intermediate Metabolizer (IM) (codeine /...

codeine antidepressants

WHAT WAS FOUND

Genotype predicts altered handling of codeine antidepressants.

WHY IT MATTERS

CYP2D6 activity governs conversion of codeine to morphine and clearance of several antidepressants, so one dose can be ineffective in some people and excessive in others.

WHAT COULD CHANGE

Avoid opioid toxicity or ineffective analgesia/SSRI dosing

KEY DEPENDENCY

Value accrues only if codeine antidepressants is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

\$210

Medical cost averted	\$120
Health gain	0.001 QALY
Cost to act	\$30 registry default

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on the methods page.

SLCO1B1 Intermediate Function (IM-like) (simvastatin)

simvastatin

WHAT WAS FOUND

Genotype predicts altered handling of simvastatin.

WHY IT MATTERS

SLCO1B1 transports statins into the liver. Reduced function raises systemic exposure and with it the risk of muscle toxicity, most clearly for simvastatin.

WHAT COULD CHANGE

Prevent statin-induced myopathy / rhabdomyolysis

KEY DEPENDENCY

Value accrues only if simvastatin is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

\$90

Medical cost averted	\$60
Health gain	0.001 QALY
Cost to act	\$20 registry default

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on the methods page.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk, prevention & exploratory findings

Risk and prevention findings, in detail

The findings above carry an evidence grade rather than a single standard of proof: this page runs from well-established variants to exploratory ones, and the badge on each card says which is which. Lower confidence changes how a finding should be read, not whether it is shown. A negative expected NMB means the modelled cost of acting exceeds the modelled benefit — useful information, not an error. Evidence grade reflects the strength of the association, not the size of the value; provenance is on the methods page.

BRCA1 pathogenic variant (heterozygous)

HIGH

POTENTIAL DECISION

Enhanced screening + risk-reducing surgery option

EXPECTED NMB \$17,308

medical cost averted \$17,412

health gain 0.019 QALY

cost to act \$2,000

LDLR familial hypercholesterolemia (heterozygous)

HIGH

POTENTIAL DECISION

High-intensity statin + cascade screening

EXPECTED NMB \$9,442

medical cost averted \$9,136

health gain 0.008 QALY

cost to act \$500

APOB familial hypercholesterolemia (heterozygous)

HIGH

POTENTIAL DECISION

High-intensity statin therapy

EXPECTED NMB \$6,658

medical cost averted \$6,578

health gain 0.006 QALY

cost to act \$500

CHEK2 I157T carrier — moderate cancer risk awareness

MODERATE

POTENTIAL DECISION

Risk-aware screening schedule

EXPECTED NMB \$3,739

medical cost averted \$3,869

health gain 0.004 QALY

cost to act \$500

registry default

APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)

MODERATE

POTENTIAL DECISION

Intensive cardiovascular risk reduction (statin) to prevent MI — Intensive statin therapy + lipid monitoring

EXPECTED NMB \$3,514

medical cost averted \$3,869

health gain 0.001 QALY

cost to act \$500

registry default

Where no finding-specific caveat is shown: baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis, and neither is measured for this individual.

Reported without a standardised value (13): 2 actionable wellness variant(s), HFE C282Y carrier — reproductive..., GABRA2 — Alcohol-metabolism..., OPRM1 — Opioid-response pharmacogenomic..., CHRNA5 — Nicotine-dependence..., CRHR1 — Stress-axis variant..., and 7 more. No registry-backed economic pathway exists for these yet, so none is invented. Each appears with its decision and evidence grade on the glance pages.

How the findings combine

Taken together, what do these findings mean?

This is the canonical reference case. Findings bearing on the same condition are combined on the risk scale, each condition's cost of illness is charged exactly once, and trial efficacy is discounted to real-world adherence before any value is credited.

INCREMENTAL HEALTHCARE COST

-\$27,378

healthcare-sector perspective

INCREMENTAL QALYS

0.063

about 23 days of healthy life on average

REFERENCE-CASE NMB

\$33,637

at \$100,000 per QALY

CLASSIFICATION

dominant (more health, lower cost)

not defined — strategy dominates or is dominated

Condition pools — combined before value is credited

CONDITION	FINDINGS	BASELINE	RAW UNCORRECTED	POOLED	ADHERENCE	REALISED	COST AVERTED	QALYS	NMB
BreastOvarian	2	60.0%	75%	53%	65%	36%	\$16,838	0.018	\$17,372
CAD	3	50.0%	98%	60%	50%	34%	\$7,410	0.022	\$9,339
Autoimmune	1	20.0%	30%	30%	65%	20%	\$1,918	0.003	\$1,929
Pathogenic	1	20.0%	30%	30%	65%	20%	\$2,019	0.002	\$1,845
Alzheimer	1	15.0%	30%	30%	35%	11%	\$1,630	0.001	\$1,516
IronOverload	3	20.0%	32%	17%	65%	12%	\$478	0.010	\$1,148
Depression	2	20.0%	18%	13%	50%	7%	\$379	0.007	\$817
Coeliac	1	20.0%	30%	30%	65%	20%	\$283	0.0001	-\$29

Raw is an uncorrected sum, not a modelled result. It adds the risk reductions of findings that share a condition, which is not how probabilities combine. Any value above 100% is the artifact this correction removes. The pooled figure beside it is the model's answer.

Two corrections, applied in order

The engine reports these as one combined reduction. They rest on different evidence and are separated here: overlap is a structural correction, adherence a behavioural one.

NAIVE AVOIDED COST

\$69,970

every finding counted independently

POOLED, AT TRIAL EFFICACY

\$50,618

28% removed — one liability, one cost of illness

REALISED BENEFIT

\$30,953

39% further removed — efficacy is not effectiveness

Adherence is charged to the benefit *and* to ongoing intervention cost, but not to the one-off test cost. 0.041 QALYs are lost to imperfect adherence relative to trial efficacy.

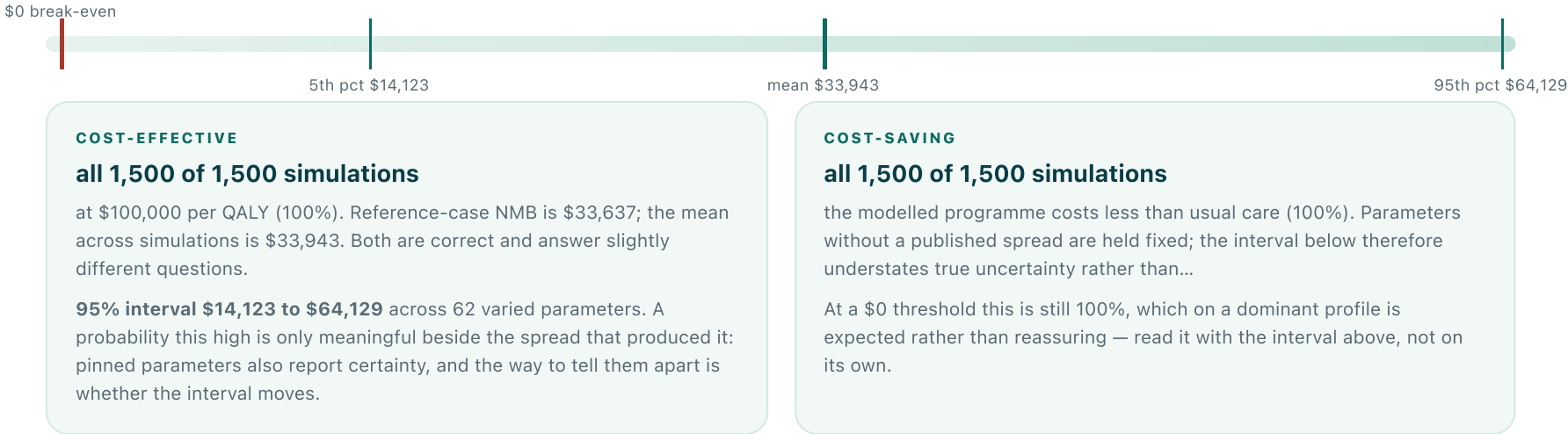
\$33,637

Uncertainty, evidence & model discipline

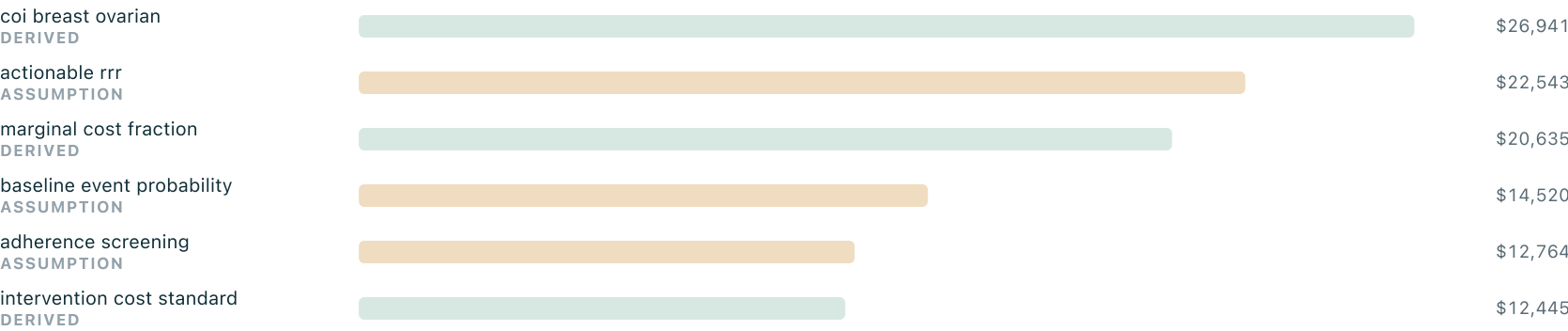
How sure is this, and what rests on judgement?

Uncertainty has two layers: the spread of modelled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

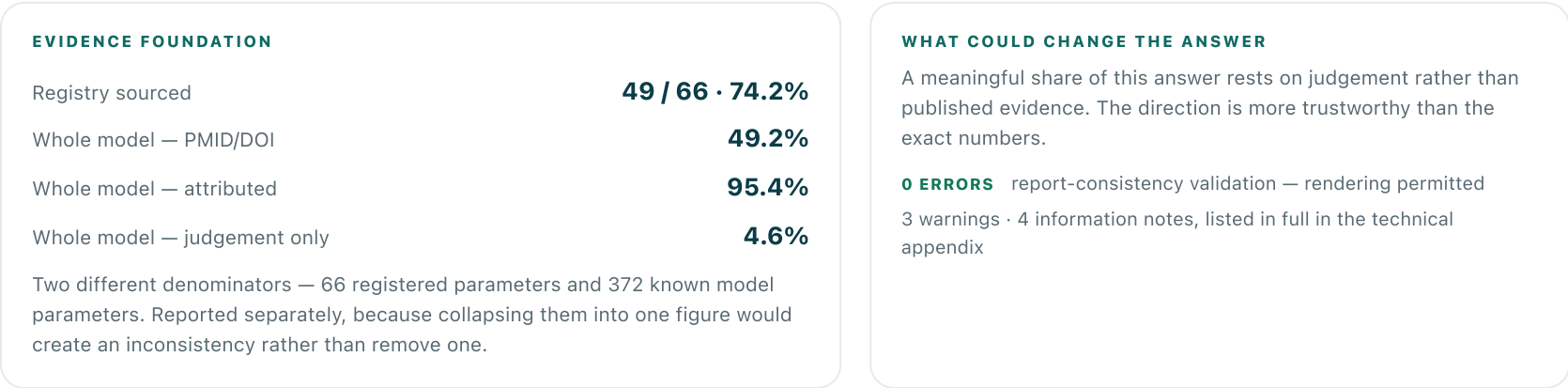
Net monetary benefit across 1,500 simulations



What drives the answer



Swing in net monetary benefit across each parameter's plausible range. Amber bars are declared assumptions rather than sourced values; they account for 47.1% of total swing.



The testing decision

Is whole-genome sequencing worth it?

Two different questions live on this page and they are never added together. One is a choice made *before* testing, averaged over a population. The other counts what sequencing actually found in this genome.

PROSPECTIVE — BEFORE TESTING		OBSERVED — IN THIS GENOME	
What is the expected value of choosing sequencing rather than an array?		What did sequencing actually find that an array could not?	
Chance an array misses an actionable finding	1 in 56	Sequencing-only findings present	0
Value if you are that person	\$25,189	Observed standalone contribution	\$0
Expected value, averaged over everyone	\$491	A count of findings this genome contains, not an expectation. For this input the population question is already settled.	
Extra cost over an array	\$200	Zero is an answer, not a gap. Every finding in this genome sits at a position an array could have typed, so none of them is reportable only by sequencing. That makes the observed contribution genuinely nil for this input — it does not mean sequencing is worthless, which is what the prospective column beside it measures.	
Net expected value	\$291	Not the same as the value of sequencing. This line counts only findings an array <i>cannot</i> report. Sequencing also recovers findings an array could have carried and did not, which is the larger effect — so the full chip-to-whole-genome difference is bigger than the figure above.	

Separation rule. The prospective figures describe the population this genome was drawn from; the observed figure describes the genome itself. They answer different decision questions, come from different inputs, and are never combined into one headline total.

Strategy frontier

STRATEGY	COST	QALYS	ICER VS PREVIOUS	STATUS
No testing	\$0	20.000	—	efficient
Genotyping chip	\$200	20.060	\$3,333/QALY	efficient
Whole-genome sequencing	\$600	20.120	\$6,667/QALY	efficient
Targeted clinical panel	\$900	20.090	—	dominated

Frontier values illustrate the testing choice for a population and are not this genome's reference case. This is an average over a population, and the distribution is extremely skewed: the great majority of people get nothing, and a small number get something that changes their care substantially. An expected value is the right...

Methods & provenance

What was assumed, where it came from, what was checked

The pages before this one tell the story; this page is their audit trail. Full parameter tables, references and the advanced analyses follow in the technical appendix.

MODEL		PROVENANCE	
Perspective	healthcare sector	Registered parameters	66
Comparator	usual care	Published	8
Outcomes	cost, QALYs, NMB	Derived	41
Time horizon	10 years	Declared assumptions	17
Discounting	3%, costs and QALYs	Registry sourced	74.2%
Willingness to pay	\$100,000/QALY	Model, resolvable	49.2%
Currency	USD, United States	Model, attributed	95.4%
Pooling	complement of products	Model, judgement only	4.6%
Adherence	benefit and ongoing cost	A parameter declared as an assumption may not cite a source — the registry refuses to load if one does.	
Uncertainty	1,500-run PSA		

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labelled as such.
- Costs are not inflation-normalised to a single price year.
- Findings without a registry-backed pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Validation

- PASS** NMB identity — $\lambda \times \Delta QALY - \Delta Cost$ reconciles to the reported total

PASS Condition cost charged once — no pool exceeds its cost of illness

PASS WGS basis separation — observed and prospective never combined

PASS Report validation gate — no errors; rendering permitted
- PASS** Dominance classification — sign of cost and effect agrees with the label

PASS Pooled risk cap — combined risk reduction stays within its bound

PASS Budget-impact consistency — peak burden is the maximum net spend

Audit trail. Every figure in this report is serialised in `economics-payload.json`, the canonical object these pages render from. The parameter registry, references, cross-path reconciliation and the advanced analyses follow in the technical appendix. Build GENOMELENS-BUILD: aa929da 2026-09-05T10:07:10-04:00. Synthetic input — no human genome was used.